Variational Regularization Network With Attentive Deep Prior for Hyperspectral–Multispectral Image Fusion

Jingxiang Yang, *Member, IEEE*, Liang Xiao, *Member, IEEE*, Yong-Qiang Zhao, *Member, IEEE*, and Jonathan Cheung-Wai Chan

***Abstract*— Hyperspectral–multispectral image (HSI-MSI) fusion relies on a robust degradation model and data prior, where the former describes the degeneration of HSI in the spectral and spatial domains, and the latter reveals the latent statistics of the expected high-resolution (HR) HSI. In practice, the degradation model is often unknown, and the data prior is usually too complicated to be expressed analytically. In this study, we propose a variational network for HSI-MSI fusion (VaFuNet), in which the degradation model and data prior are implicitly represented by a deep learning network and jointly learned from the training data. A variational fusion model regularized by deep prior is first proposed, and then, it is optimized via a half-quadratic splitting and unfolded into a deep network. The deep prior is implicitly represented by a proximity operator. Due to the structural self-similarity, HSI possesses structural recurrences across different scales. To exploit such nonlocal prior and enhance the representability of network, we also propose a multiscale nonlocal attention and embed it into the deep prior proximity. The degradation model and deep prior proximity are jointly learned via end-to-end training. Experimental results on simulated and real-life HSI datasets demonstrate the effectiveness of the proposed VaFuNet HSI-MSI fusion method.**

***Index Terms*— Attention, deep, enhancement, fusion, hyperspectral, multispectral, nonlocal.**

I. Introduction

HYPERSPECTRAL image (HSI) is captured over a certain electromagnetic range with many contiguous spectral bands at narrow bandwidth. The spectral and spatial information in HSI endows it with discriminative ability to distinguish different materials in the scene [1]. HSI has been widely applied in many fields, including land-cover classification [2], mineral exploration [3], and target detection [4]. Due to the narrow bandwidth, the photonic energy in hyperspectral imaging is limited, and the resolution of HSI is often moderate, which prevents its applications that require high-resolution (HR) data such as subpixel target detection [5] and fine urban mapping [6].

Multispectral image (MSI) often has wider bandwidth and finer resolution. The complementary information with detailed surface structures in MSI can be used for enhancing the low-resolution (LR) HSI. With increasing availability of multispectral imaging missions, it is possible to collect HR MSI over the same scene in a similar period as the LR HSI. Actually, HSI-MSI fusion has become an important technology for HSI resolution enhancement where the LR HSI is super-resolved to the spatial resolution of MSI after fusion [7].

Degradation model and data prior are two key aspects in HSI-MSI fusion. To infer the HR HSI, the degradation relationship between the expected HR HSI and the observed LR HSI and HR MSI should be explored. However, in practice, the degradation model is often unknown. On the other hand, the data prior reveals the latent statistics of the HR HSI and regularizes the ill-posed HSI-MSI fusion problem. Since both spectral information and spatial information in HSI should be characterized, it is challenging to design a handcrafted prior that fully captures the cross-modality statistics of the spatial and spectral domains. In fact, the true data prior of HSI is complicated to express using a handcrafted prior [8].

To address these issues, in this study, we propose a variational network for HSI-MSI fusion (VaFuNet), in which the degradation model and data prior are implicitly represented by a deep learning (DL) network and jointly learned from the training data. We first design a variational HSI-MSI fusion model regularized by deep prior. Then, we optimize the fusion model via half-quadratic splitting and alternative optimization. The VaFuNet is built by unfolding the iterative optimization phases into a deep network. The deep prior is implicitly expressed by a proximity network and learned from external training data. Due to the self-similarity property, the HSI possesses structural recurrences across different scales. To utilize such prior and enhance the representability of network, we propose a multiscale nonlocal attention and embed it into the proximity network to represent the deep

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Jingxiang Yang and Liang Xiao are with the School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing 210094, China (e-mail: yang123jx@njust.edu.cn; xiaoliang@njust.edu.cn).

Yong-Qiang Zhao is with the School of Automation, Northwestern Polytechnical University, Xi'an 710129, China (e-mail: zhaoyq@nwpu.edu.cn).

Jonathan Cheung-Wai Chan is with the Department of Electronics and Informatics, Vrije Universiteit Brussel, 1050 Brussels, Belgium (e-mail: jcheungw@etrovub.be).

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prior, which is jointly learned with the degradation model via end-to-end training. The experimental results on several HSI datasets demonstrate that the proposed method is competitive by comparing it with other state-of-the-art HSI-MSI fusion methods.

The proposed fusion method makes three contributions.

1) We propose a deep prior regularized variational HSI-MSI fusion model where its optimization is obtained by unfolding into a deep network called VaFuNet. The degradation model and deep prior regularization are implicitly expressed by the network and jointly learned from the training data in an end-to-end manner.
2) To further exploit the self-similarity prior of HSI, we propose a multiscale nonlocal attention mechanism, in which the long-range interdependencies across different scales of HSI can be exploited. By embedding the multiscale nonlocal attention into the deep prior proximity, the representability of the network is enhanced.
3) We propose to jointly utilize the deep prior and the multiscale nonlocal prior of HSI for HSI-MSI fusion. The rationality behind integrating deep and multiscale nonlocal priors for HSI-MSI fusion is demonstrated by the experiments.

The remainder of this article is organized as follows. In Section II, we introduce the background related to our work. In Section III, the proposed VaFuNet is presented, including the deep variational fusion model, the multiscale nonlocal attentive proximity, and the network implementation. The experiments and discussion are given in Section IV. We state the conclusions in Section V.

II. Related Works

A. HSI-MSI Fusion

The HSI-MSI fusion technology can be roughly separated into two categories: model-driven fusion methods and learning-driven fusion methods.

1) Model-Driven HSI-MSI Fusion: Model-driven HSI-MSI fusion methods estimate the HR HSI based on the degradation model and prior knowledge. The degradation model describes the degeneration of HR HSI, and it depends on the point spread function (PSF) and spectral response function (SRF) of sensors. The prior knowledge reveals the statistics of the latent HR HSI. In a variational fusion model, the degradation and prior act as the data fidelity and regularization, respectively. Based on the statistics of latent HSI, several handcrafted priors have been designed. For example, a subspace sparsity prior of latent HSI was designed for HSI-MSI fusion in [9]. Dian *et al.* [10] proposed to exploit sparsity and structural self-similarity priors for HSI-MSI fusion. HSI possesses a low-rank characteristic due to its spectral correlation [2], [11], and the low-rank prior could be used together with spectral manifold embedding and graph regularization for HSI-MSI fusion [12], [13]. It is shown that total variation (TV) prior can promote gradient consistency. In [14], an HSI-MSI fusion method was proposed with a vector TV prior that could align the edges and details along the spectral domain. In [15], hybrid unidirectional and anisotropic TV priors were exploited to enhance the robustness of the reconstructed HSI over mixed noise. It is noted that the degradation model (i.e., PSF and SRF) should be given or accurately estimated in advance in these model-driven methods.

HSI-MSI fusion can also be conducted according to the linear spectral mixture model (LMM), in which HSI and MSI are unmixed into endmembers and abundances. For example, based on the LMM, Yokoya *et al.* [16] proposed a coupled nonnegative matrix factorization (CNMF) method, the LR HSI and HR MSI were alternatively unmixed in a coupled manner, and then, the HR HSI was produced with the endmembers of LR HSI and abundances of MSI. Similarly, by introducing a sparsity constraint on the endmembers, a sparse matrix factorization-based HSI-MSI fusion method was proposed in [17]. Both of the methods in [16] and [17] require the spectral degradation model. In [18], the LR HSI was first enhanced through single-frame super-resolution with a dedicated PSF, and then, it is fused with the HR MSI via hybrid color mapping (HCM) to generate the HR HSI.

The above HSI-MSI fusion methods are designed with the degradation and expert prior on latent HSI, and these models are mathematically interpretable. However, model-driven methods usually involve a time-consuming iterative optimization process. Furthermore, some complex spectral–spatial statistics of latent HSI may not be explicitly expressed by the handcrafted priors, which limits the performance of model-driven fusion methods.

2) Learning-Driven HSI-MSI Fusion: By learning a DL network that represents the mapping between degraded and desired images, the HR HSI can be inferred from its LR version. Several DL-based HSI-MSI fusion methods have been proposed. For example, a 3-D convolutional neural network (CNN) was proposed to learn the mapping between stacked LR HSI and HR MSI with HR HSI in [19]. In [20], a two-branch CNN (Two-CNN) was proposed for HSI-MSI fusion, where each branch extracted the features individually from LR HSI and HR MSI, and then, the HR HSI was inferred by fusing the features with fully connected (FC) layers. In [21], a pyramid fusion CNN was proposed to extract multiscale features from HR MSI and then progressively super-resolve the LR HSI. The above DL networks for HSI-MSI fusion directly learn a brute-force mapping between the LR and HR HSIs, and they are black-boxes with limited interpretability. In addition, the spatial and spectral degradation information is often neglected and cannot be exploited in these DL networks.

Some learning-driven HSI-MSI fusion methods have tried to employ the degradation model with the DL networks to improve the fusion performance. Dian *et al.* [22] integrated the known degradation model with DL in the Sylvester framework for HSI sharpening. In [23], with known spectral degradation, the spatial degradation model was learned in a DL network, and then, adaption learning was used to produce the HR HSI. In [24], a fusion network containing degradation estimation modules and fusion modules was proposed, and the spatial and spectral degradations were learned and then employed for HSI-MSI fusion. Xie *et al.* [25] proposed to learn the spatial degradation in a DL network, and the spectral correlation of latent HSI was considered via the low-rank constraint.

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It was shown that the fusion performance can be significantly improved by exploiting the degradation information in DL fusion networks.

Unsupervised DL-based HSI-MSI fusion is another important approach, which avoids the requirement for large training data. Unsuprevised DL-based networks often contain encoder–decoder structures, where unmixed abundance and endmembers were represented [26]. To improve the spatial details extraction and injection, self-attention was introduced into the unsupervised fusion network in [27]. In [28], an unsupervised DL fusion network was proposed for unregistered HSI and MSI by projecting them into the same statistical space and maximizing the mutual information.

The learning-driven methods learn the prior automatically from the data; comparatively, the model-driven methods use handcrafted prior and can easily exploit the degradation model in fusion. How the DL-based learning-driven methods could learn the unknown degradations in both spatial and spectral domains has not been thoroughly investigated. In addition, the interpretability of the learning-driven method needs to be enhanced, and integrating the model-driven method with it may be one direction, which also lacks study.

B. Attention Mechanism

Attention mechanism originates from the human visual system and makes the DL networks concentrate on the important regions in the scene [29]. By embedding attention learning into a DL network, the features are recalibrated to emphasize the informative features while suppressing the unimportant ones. The representability of a network can be enhanced with the attention mechanism. Several attention mechanisms have been proposed recently. Hu *et al.* [30] proposed a channel-wise attention to recalibrate different feature channels using FC layers, and informative channels were recalibrated with higher weights. Woo *et al.* [31] proposed a spatial attention mechanism using convolutional layers, and the informative regions in each feature map can be indicated by the learned attention map. A lightweight version of [31] was proposed in [32], in which channel and spatial attention maps were produced by the FC layers and the dilated convolutional layers, respectively.

Most of the current methods learn attention maps with convolutional layers. Due to the limited receptive field, only local contextual information can be exploited. If the long-range interdependency in HSI could be employed for attention learning, the representability of DL would be boosted.

III. Deep Variational HSI-MSI Fusion Network

A. Framework of Deep HSI-MSI Fusion

HSI-MSI fusion aims to generate an expected HR HSI given a pair of LR HSI and HR MSI taken from the same scene. For the sake of expression simplicity, we denote the expected HR HSI as a matrix $\mathbf{X} \in \mathbb{R}^{\mathrm{MN}\times L}$, where M, N, and L are the height, width, and spectral bands of $\mathbf{X}$, respectively. The degradation model among $\mathbf{X}$, LR HSI, and HR MSI can be expressed as

$$\mathbf{Y} = \mathbf{CX} \tag{1}$$

$$\mathbf{Z} = \mathbf{XR} \tag{2}$$

where $\mathbf{Y} \in \mathbb{R}^{\mathrm{mn}\times L}$ denotes the observed LR HSI with m, n, and L as its height, width, and spectral bands, respectively, and $(m < M, n < N)$, $r = (M/m) = (N/n)$ is the upscaling factor of HSI-MSI fusion. $\mathbf{Z} \in \mathbb{R}^{\mathrm{MN}\times l}$ is the observed HR MSI with M, N, and l as its height, width, and spectral bands, respectively $(l < L)$. $\mathbf{C} \in \mathbb{R}^{\mathrm{mn}\times \mathrm{MN}}$ denotes the spatial degradation, which is the composition of a convolution between the PSF with $\mathbf{X}$ and a spatial downsampling. $\mathbf{R} \in \mathbb{R}^{L\times l}$ denotes the spectral degradation determined by the SRFs of the sensors. It is noted that both of $\mathbf{C}$ and $\mathbf{R}$ are often unknown and should be estimated [14].

With the premise that both $\mathbf{C}$ and $\mathbf{R}$ are given, the HR HSI $\mathbf{X}$ can be reconstructed in the variational HSI-MSI fusion framework by solving the objective function

$$\hat{\mathbf{X}} = \underset{\mathbf{X}}{\operatorname{argmin}} ||\mathbf{Y} - \mathbf{CX}||_F^2 + \mu||\mathbf{Z} - \mathbf{XR}||_F^2 + \lambda J(\mathbf{X}) \tag{3}$$

where $||.||_F$ is the Frobenius norm used to measure the data fidelity and $J(.)$ is a regularization term associated with the prior knowledge of latent $\mathbf{X}$. Parameters μ and λ balance the tradeoff between data fidelity and regularization. In this study, inspired by the success of DL, we propose to use data prior J_{deep} that is represented by the DL network and learned from the training data to regularize the HSI-MSI fusion.

The objective function (3) can be optimized with the half-quadratic splitting method [33], [34]. After introducing an auxiliary variable $\mathbf{V}$, (3) can be written as

$$\hat{\mathbf{X}} = \underset{\mathbf{X}}{\operatorname{argmin}} ||\mathbf{Y} - \mathbf{CX}||_F^2 + \mu||\mathbf{Z} - \mathbf{XR}||_F^2 + \lambda J_{\mathrm{deep}}(\mathbf{V}) + \rho||\mathbf{X} - \mathbf{V}||_F^2 \tag{4}$$

where ρ is the weight of the quadratic penalty. With alternative optimization method, (4) can be divided into two subproblems with updating rules at the $(k+1)$th iteration phase as follows.

1) X-Subproblem:

$$\mathbf{X}^{k+1} = \underset{\mathbf{X}}{\operatorname{argmin}} ||\mathbf{Y} - \mathbf{CX}||_F^2 + \mu||\mathbf{Z} - \mathbf{XR}||_F^2 + \rho||\mathbf{X} - \mathbf{V}^k||_F^2. \tag{5}$$

The $\mathbf{X}$-subproblem has a closed-form solution

$$\mathrm{vec}(\mathbf{X}^{k+1}) = \left[\mathbf{I}_L \otimes (\mathbf{C}^T\mathbf{C} + \rho\mathbf{I}_{\mathrm{MN}}) + (\mu\mathbf{R}\mathbf{R}^T) \otimes \mathbf{I}_{\mathrm{MN}}\right]^{-1} \times \mathrm{vec}(\mathbf{C}^T\mathbf{Y} + \mu\mathbf{Z}\mathbf{R}^T + \rho\mathbf{V}^k) \tag{6}$$

where vec(.) denotes vectorization, $\otimes$ is the Kronecker product, and $\mathbf{I}_L$ and $\mathbf{I}_{\mathrm{MN}}$ are identity matrices with sizes $L \times L$ and $MN \times MN$, respectively. However, solving (6) incurs a high computational cost because it involves an inversion of a huge matrix with size $MNL \times MNL$. Instead of the closed-form solution, we update $\mathbf{X}^{k+1}$ using the iterative gradient descent optimization method

$$\mathbf{X}^{k+1} = \mathbf{X}^k + \delta\big[-\mathbf{C}^T\mathbf{Y} + \mathbf{C}^T\mathbf{C}\mathbf{X}^k - \mu\mathbf{Z}\mathbf{R}^T + \mu\mathbf{X}^k\mathbf{R}\mathbf{R}^T - \rho\mathbf{V}^k + \rho\mathbf{X}^k\big] \tag{7}$$

where δ is the step size of gradient descent. It is noted that multiple iterative steps are required in the gradient descent optimization of (5); here, for simplicity, we set the single step of gradient descent to approximate $\mathbf{X}^{k+1}$, which facilitates the network implementation [33].

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2) V-Subproblem:

$$\mathbf{V}^{k+1} = \underset{\mathbf{V}}{\operatorname{argmin}} \frac{1}{2}||\mathbf{X}^{k+1} - \mathbf{V}||_F^2 + \frac{\lambda}{2\rho} J_{\text{deep}}(\mathbf{V}). \tag{8}$$

The **V**-subproblem can also be formulated as $\mathbf{V}^{k+1} = \text{prox}_{(\lambda/2\rho)J}(\mathbf{X}^{k+1})$ denoting a proximity operator of $\lambda J_{\text{deep}}/2\rho$ at point $\mathbf{X}^{k+1}$. Since the deep prior J_{deep} has no explicit mathematical expression, it is difficult to obtain the closed-form solution of the **V**-subproblem. On the other hand, $\mathbf{V}^{k+1}$ can be treated as the "denoised" version of $\mathbf{X}^{k+1}$, and it can be approximated using any denoising algorithm in a plug-and-play manner with $\mathbf{X}^{k+1}$ as the "noisy" input. In this study, a multiscale nonlocal attentive CNN denoiser is designed as the proximity operator to solve (8), which will be introduced in Section III-C.

B. Network Implementation

The HR HSI can be reconstructed via iteratively solving the above **X**- and **V**-subproblems until the convergence to a local minimum is achieved; however, such an iterative process is time-consuming. Instead, we propose VaFuNet via unfolding the iterations and executing each iterative phase with the DL network, the HR HSI can be inferred through a feedforward computation of VaFuNet. The degradation model and data prior can be implicitly represented in VaFuNet. The deep prior, the degradation model parameters, and hyperparameters are automatically learned from the training data in an end-to-end fashion.

1) Iterations Unfolding: To facilitate the network design, in this section, both HSI and MSI are represented in the tensor format rather than in the matrix format. Given LR HSI $\mathcal{Y} \in \mathbb{R}^{m\times n\times L}$ and HR MSI $\mathcal{Z} \in \mathbb{R}^{M\times N\times l}$, the **X**-subproblem in (7) can be rewritten in the tensor format as

$$\begin{aligned}\mathcal{X}^{k+1} = \mathcal{X}^k + \delta\big[&- \mathcal{C}^T\mathcal{Y} + \mathcal{C}^T\mathcal{C}\mathcal{X}^k - \mu\mathcal{Z}\times_3\mathbf{R} \\ &+ \mu\mathcal{X}^k\times_3\mathbf{R}^T\times_3\mathbf{R} - \rho\mathcal{V}^k + \rho\mathcal{X}^k\big] \end{aligned}\tag{9}$$

where $\mathcal{X}^{k+1} \in \mathbb{R}^{M\times N\times L}$ is the intermediate HR HSI at the $(k+1)$th iterative phase and $\times_3$ is mode-3 tensor multiplication. Specifically, the symbols $\mathcal{C}$ and $\mathcal{C}^T$ are introduced as spatial degradation and its inverse operator on the tensor, respectively, which have the same effect as **C** and $\mathbf{C}^T$ in (7). They can be approximated via the convolutional and transposed convolutional layers, respectively, with kernel sizes and strides set to the upscaling factor r [25]. Mode-3 tensor multiplication with $\mathbf{R}^T$ represents spectral degradation, and it can be executed via the convolutional layer with kernel size 1×1 and number of features l. Similarly, mode-3 tensor multiplication with **R** can be performed via the convolutional layer with kernel size 1×1 and number of features L.

The **V**-subproblem at the $(k+1)$th iteration phase can also be rewritten in tensor format as

$$\mathcal{V}^{k+1} = \text{prox}_{\frac{\lambda}{2\rho}J}\left(\mathcal{X}^{k+1}\right) \tag{10}$$

where $\mathcal{V}^{k+1} \in \mathbb{R}^{M\times N\times L}$ is the auxiliary variable. We will describe the network architecture of the proximity operator in Section III-B3.

2) Iterations Initialization: The initialization of the estimated HR HSI at the first iterative phase is performed by simply fusing the spatially upsampled $\mathcal{Y}$ and spectrally upsampled $\mathcal{Z}$, which is formulated as

$$\mathcal{X}^0 = \left(\mathcal{C}^T\mathcal{Y} + \mathcal{Z}\times_3\mathbf{R}\right)/2 \tag{11}$$

where $\mathcal{X}^0 \in \mathbb{R}^{M\times N\times L}$ is the initialized HR HSI. The spatial upsampling is performed via the transposed convolutional layer, and the kernel size and stride are set to the upscaling factor r. The spectral upsampling is performed via the convolutional layer with kernel size 1×1 and number of features L.

By unfolding (9)–(11), we build the VaFuNet with multiple modules corresponding to the iterative phases, in which the **X**- and **V**- subproblems are executed, as shown in Fig. 1. The network parameters are composed of the degradation parameters (e.g., $\mathcal{C}$ and **R**), hyperparameters (e.g., μ and ρ), and deep prior proximity in (9)–(11). Through training, the VaFuNet jointly learns the degradation parameters, hyperparamters, and the deep prior proximity in an end-to-end manner. It is noted that we allow the degradation parameters to vary across modules because the fusion quality does change with different modules.

C. Multiscale Nonlocal Attentive CNN Proximity

The solution of the **V**-subproblem in (10) is a proximity operator that recovers $\mathcal{V}^{k+1}$ from $\mathcal{X}^{k+1}$. Any off-the-shelf denoising DL network can be adopted here for solving the **V**-subproblem. To enhance the DL representability for the data prior and exploit the structural self-similarity prior of HSI, we propose a multiscale nonlocal attentive CNN proximity operator. In this section, we first introduce two attention modules: the channel-wise attention module and the local spatial attention modules [30], [31]. Then, we propose a multiscale nonlocal spatial attention module. By embedding it into a residual CNN, we finally obtain the multiscale nonlocal attentive CNN proximity operator.

1) Channel-Wise Attention: The channel-wise attention selects the informative channels by exploiting the interdependencies among the different feature channels. For an intermediate feature $\mathcal{F} \in \mathbb{R}^{W\times H\times C}$, W and H are the spatial sizes, and C is the number of feature channels. Global average pooling and max pooling are applied to each channel of $\mathcal{F}$ to aggregate the information in the spatial domain, and the pooled features can be treated as indicators of the importance of each channel. The pooled features are then fed to two FC layers for channel-wise attention [31]:

$$\begin{aligned}\mathcal{A}_C = \sigma\big[&\mathbf{W}_1(\mathbf{W}_0\cdot\text{glb_avgpool}(\mathcal{F})) \\ &+ \mathbf{W}_1(\mathbf{W}_0\cdot\text{glb_maxpool}(\mathcal{F}))\big]\end{aligned}\tag{12}$$

where $\mathbf{W}_0 \in \mathbb{R}^{C/s\times C}$ and $\mathbf{W}_1 \in \mathbb{R}^{C\times C/s}$ are the weight matrices of the FC layers, s is a ratio to reduce the computational burden of the FC layers ($s = 4$ in our implementation), and $\sigma(.)$ is the sigmoid activation function that maps the attention values into the range of [0, 1]. The channel-wise attention map $\mathcal{A}_C \in \mathbb{R}^{C\times 1}$ reflects the importance of different channels.

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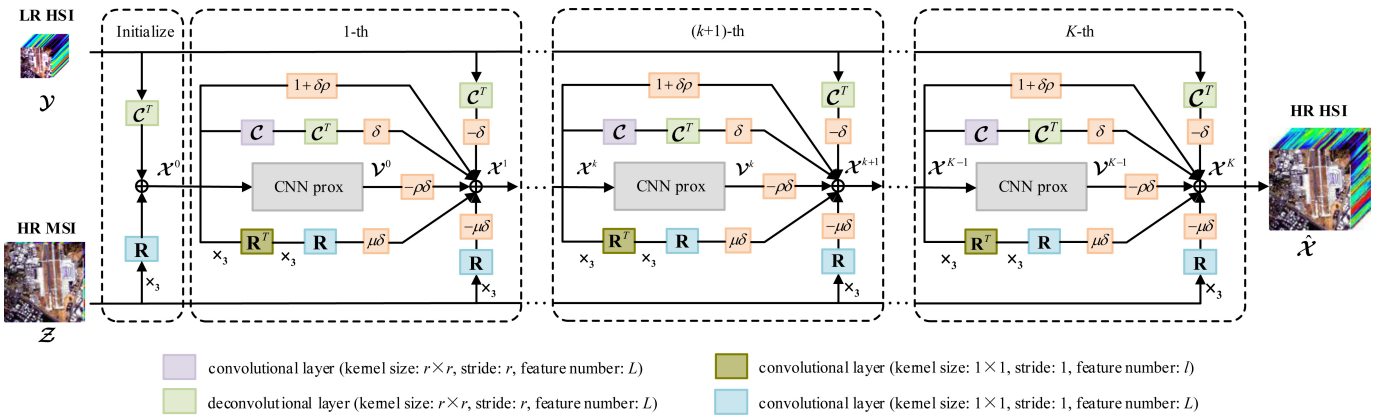


Fig. 1. Network architecture of VaFuNet for HSI-MSI fusion.

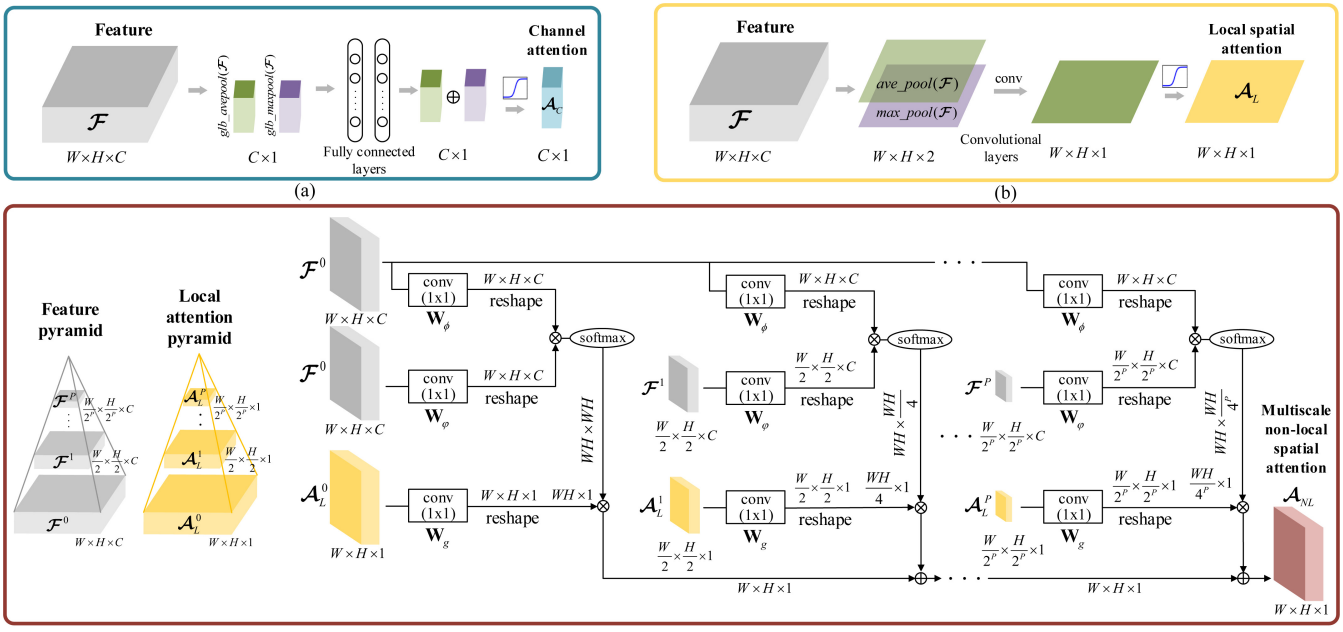


Fig. 2. Illustrations of attention mechanisms. (a) Channel attention. (b) Local spatial attention. (c) Multiscale nonlocal attention.

2) Local Spatial Attention: The spatial attention map indicates and emphasizes informative regions in each feature channel. To reduce the data size and computational burden, average pooling and max pooling are applied along the channel dimension of $\mathcal{F}$, two pooled features with size $W \times H$ would be obtained, and then, they are concatenated and fed to a convolutional layer with kernel size 3×3 for the spatial attention map [31]

$$\mathcal{A}_L = \sigma\{\text{conv}[\text{avg_pool}(\mathcal{F}); \text{max_pool}(\mathcal{F})]\} \tag{13}$$

where $\mathcal{A}_L \in \mathbb{R}^{W\times H\times 1}$ is the spatial attention map, which reflects the importance of different spatial regions. The illustration of (13) is shown in Fig. 2(b). Due to the limited receptive field of convolutional layer, only local contextual information can be aggregated and encoded for spatial attention learning, which limits the representability of attention.

3) Multiscale Nonlocal Spatial Attention: HSI possesses structural recurrences across different scales, and exploiting such multiscale long-range interdependencies in attention learning will boost the representability of the data prior. Inspired by the nonlocal network [35], we propose a multiscale nonlocal spatial attention mechanism, in which the long-range interdependencies can be aggregated from the multiscale spaces of the features in attention learning. A pyramid of feature $\mathcal{F}$ is first constructed as $\{\mathcal{F}^0, \mathcal{F}^1, \ldots, \mathcal{F}^P\}$

$$\mathcal{F}^P = h(\mathcal{F}^{P-1}) = h(h(\mathcal{F}^{P-2})) = h, \ldots, (h(\mathcal{F}^0)) \tag{14}$$

where $\mathcal{F}^0 = \mathcal{F}$, P is the number of levels of pyramid, and $h(.)$ denotes a convolutional layer with stride 2 used to produce the coarser-scale feature. It should be noted that rather than using downsampling, we use the convolutional layer $h(.)$ to generate the coarser-scale features, which could reduce the information loss caused by downsampling. Similarly, an attention pyramid $\{\mathcal{A}_L^0, \mathcal{A}_L^1, \ldots, \mathcal{A}_L^P\}$ can be obtained from the local spatial attention $\mathcal{A}_L$. We measure the pairwise correlation between the pixels across different scales in the feature pyramid. With the

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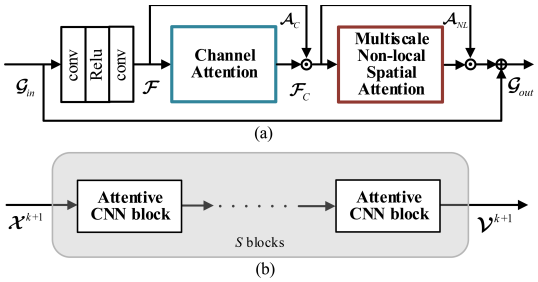


Fig. 3. (a) Attentive CNN block with channel and multiscale nonlocal attention embedded. (b) Attentive CNN proximity composed of multiple attentive CNN blocks in (a).

pairwise correlation, the multiscale nonlocal attention is computed as the weighted average of the local attention at all pixels in the pyramid $\{\mathcal{A}_L^0, \mathcal{A}_L^1, \ldots, \mathcal{A}_L^P\}$, which is formulated as

$$\mathcal{A}_{\mathrm{NL}(i)} = \frac{1}{H} \sum_p \sum_j f\left(\mathcal{F}_{(i)}^0, \mathcal{F}_{(j)}^p\right) g\left(\mathcal{A}_{L}{}_{(j)}^{p}\right) \tag{15}$$

where $\mathcal{A}_{\mathrm{NL}(i)}$ is the multiscale nonlocal attention values at the ith pixel, $g(\mathcal{A}_{L}{}_{(j)}^{p}) = \mathbf{W}_g \mathcal{A}_{L}{}_{(j)}^{p}$ is the representation of the local spatial attention at the jth pixel, f measures the correlation between the ith and jth pixels in the feature pyramid, and $H = \sum_p \sum_j \sum_i f(\mathcal{F}_{(i)}^0, \mathcal{F}_{(j)}^p)$ is a constant for normalization. The pairwise correlation is measured in the Gaussian space [35]

$$f\left(\mathcal{F}_{(i)}^0, \mathcal{F}_{(j)}^p\right) = \exp\left(\phi\left(\mathcal{F}_{(i)}^0\right)^T \varphi\left(\mathcal{F}_{(j)}^p\right)\right) \tag{16}$$

where $\phi(\mathcal{F}_{(i)}^0) = \mathbf{W}_\phi \mathcal{F}_{(i)}^0$ and $\varphi(\mathcal{F}_{(j)}^p) = \mathbf{W}_\varphi \mathcal{F}_{(j)}^p$ project features $\mathcal{F}_{(i)}^0$ and $\mathcal{F}_{(j)}^p$ into a transformation space, in which the correlation is computed. The Gaussian function is implemented with the *softmax* operator, and $g(.)$, $\phi(.)$, and $\varphi(.)$ can be implemented via reshaping and convolution with kernel size 1×1, as shown in Fig. 2(c). The matrices $\mathbf{W}_g$, $\mathbf{W}_\phi$, and $\mathbf{W}_\varphi$ are trained together with other parameters in an end-to-end fashion.

4) Channel-Spatial Attentive CNN Proximity: An attentive CNN block can be constructed by embedding the channel and multiscale nonlocal spatial attention into a residual CNN, as shown in Fig. 3(a). Suppose that the input of the attentive CNN block is $\mathcal{G}_{\mathrm{in}}$, and then, deep features $\mathcal{F}$ are extracted from it by two convolutional layers. With the extracted feature $\mathcal{F}$, the channel and spatial attention maps are then learned. The output of the attentive CNN is expressed as

$$\mathcal{G}_{\mathrm{out}} = \mathcal{G}_{\mathrm{in}} + \mathcal{F}_C \odot \mathcal{A}_{\mathrm{NL}} \tag{17}$$

where $\mathcal{F}_C = \mathcal{F} \odot \mathcal{A}_C$, $\mathcal{F} = \mathrm{conv}(\mathrm{Relu}(\mathrm{conv}(\mathcal{G}_{\mathrm{in}})))$ and $\odot$ is element-wise multiplication. Since $\mathcal{A}_C$ is of size $C \times 1$, we need to pad the elements of $\mathcal{A}_C$ to match its size with $\mathcal{F}$ prior to the element-wise multiplication. Similar to [31], the channel and multiscale nonlocal spatial attention modules are arranged sequentially.

Suppose that there are S residual attentive CNN blocks constituting the CNN proximity operator. With more residual attentive CNN blocks, better performance would be obtained, but the network complexity would increase, and more training time is required. In the implementation, we empirically set one attentive CNN block (i.e., $S = 1$) in each CNN proximity, which is demonstrated to obtain the satisfactory performance. In this special case, the CNN proximity for the $\mathbf{V}$-subproblem at the $(k+1)$th iteration phase can be formulated as

$$\mathcal{V}^{k+1} = \mathcal{X}^{k+1} + \mathcal{F}_C \odot \mathcal{A}_{\mathrm{NL}} \tag{18}$$

where $\mathcal{F}_C = \mathcal{F} \odot \mathcal{A}_C$ and $\mathcal{F} = \mathrm{conv}(\mathrm{Relu}(\mathrm{conv}(\mathcal{X}^{k+1})))$.

D. Implementation Details of Training

All of the parameters in the VaFuNet network need to be trained. The L1 norm of the reconstruction errors of HR HSI is used as the loss function

$$\mathrm{loss} = \frac{1}{T} \sum_t ||\mathcal{X}^{(t)} - \mathrm{VaFuNet}(\mathcal{Y}^{(t)}, \mathcal{Z}^{(t)})||_1 \tag{19}$$

where $\mathcal{X}^{(t)}$ is the tth ground-truth HR HSI patch, $\mathcal{Y}^{(t)}$ and $\mathcal{Z}^{(t)}$ are the LR HSI and HR MSI patches, respectively, $t = 1, 2, \ldots, T$, and T is the number of training samples. The loss function is minimized using the adaptive moment estimation (ADAM) optimizer [36].

It is noted that sufficient paired images patches $\{\mathcal{X}^{(t)}, (\mathcal{Y}^{(t)}, \mathcal{Z}^{(t)})\}$ should be extracted as training samples. However, in real applications, the ground-truth HR HSI never exists, and it is not realistic to extract $\mathcal{X}^{(t)}$ directly. We adopt the strategy in [25] to create training samples following Wald protocol [37]. The DL network can be trained on resolution-reduced data and then applied on the observed LR HSI and HR MSI to produce an HR HSI.

IV. Experimental Results

In this section, the effectiveness of the proposed VaFuNet is demonstrated by comparing with other state-of-the-art methods on both simulated and real spaceborne data, and then, the sensitivity analysis of the network parameters is presented. We also analyze the reasonability of multiscale nonlocal attention for HSI-MSI fusion.

A. Experiment Setting

Three HSI datasets are used in the simulation experiment. The first one is the Interdisciplinary Computational Vision Lab (ICVL) dataset acquired by the Specim Kappa DX4 hyperspectral camera [39]. The ICVL data has 31 spectral bands in the range of 400–700 nm. The spatial size is 1300×1392. The second data are the Harvard dataset collected by the Nuance hyperspectral camera [40]. This dataset has 50 images taken in the spectral range of 420–720 nm with 31 bands. The spatial size is 1040×1392. The third data are the 2018 IEEE Geoscience and Remote Sensing Society (GRSS) Data Fusion Contest data [41] obtained by the National Center for Airborne Laser Mapping (NCALM) over Houston University in 2017. The data size is 1202×4172 with a spatial resolution of 1 m. There are 48 bands in the spectral range 380–1050 nm.

Simulation experiments are performed on the above datasets. Fifty images are taken from the Harvard and ICVL datasets, and 35 images are used for training and 15 images for blind test. The testing images are shown in Fig. 4. We follow the widely used strategy of Wald protocol [42]

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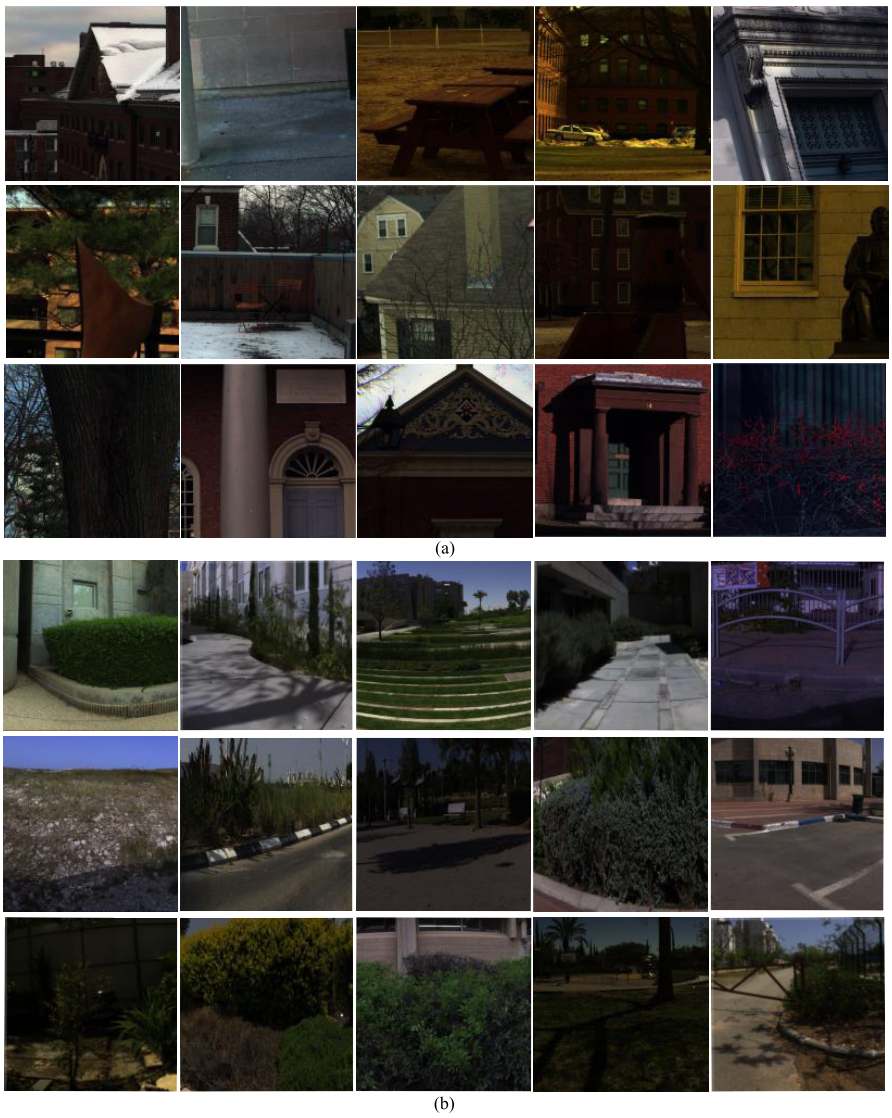


Fig. 4. Testing images selected from (a) Harvard and (b) ICVL datasets.

using image degradation for quality assessment of spatial enhancement methods [43]. The original HSIs are regarded as ground truth and used to simulate the LR HSIs and HR MSIs. The LR HSIs are simulated via Gaussian downsampling [7]. The HS-MS-Fusion toolbox [7] is adopted to simulate the LR HSI, where the size, mean, and standard deviation σ of Gaussian kernel are set to r, 0, and $r/2.3548$, respectively. The LR-HSI is produced by applying an $r \times r$ Gaussian kernel and then downsampling every r pixels in both vertical and horizontal directions. For ICVL and Harvard datasets, the HR MSIs are simulated via spectral degradation with SRF of Nikon D700 sensor [22], and the spectral bands are in the range of 440–470, 520–550, and 600–670 nm. For *Houston University*, the HR MSI is simulated through spectral degradation with SRF of Worldview-4 sensor [44], and its spectral bands are in the range of 655–690, 510–580, 450–510, and 780–920 nm.

In the training stage, paired patches are extracted from the LR HSI, HR MSI, and HR HSI as training samples. Data augmentation (rotating and flipping) [45] is adopted here to increase the volume of training set and avoid overfitting. Totally, we collect about 200 000 training samples. Experiments for spatial enhancement in the factors of 4 and 8 are conducted.

For ICVL and Harvard, training samples are extracted only from the 35 preselected training images. In case of upscaling factor 4, the sizes of LR HSI, HR MSI, and HR HSI training patches are $10 \times 10 \times 31$, $40 \times 40 \times 3$, and $40 \times 40 \times 31$, respectively; in case of upscaling factor 8, the sizes of LR HSI, HR MSI, and HR HSI training patches are $5 \times 5 \times 31$, $40 \times 40 \times 3$, and $40 \times 40 \times 31$, respectively. The trained model will be used to reconstruct the enhanced/fused image for each of the 15 testing images. The means of assessment indices on the 15 testing images are given in Table I.

For *Houston University*, a subimage with size 512×512 is cropped and set aside as a blind test. The remainder area is used for extracting training samples. There is no overlapping between the training and testing data. In case of upscaling

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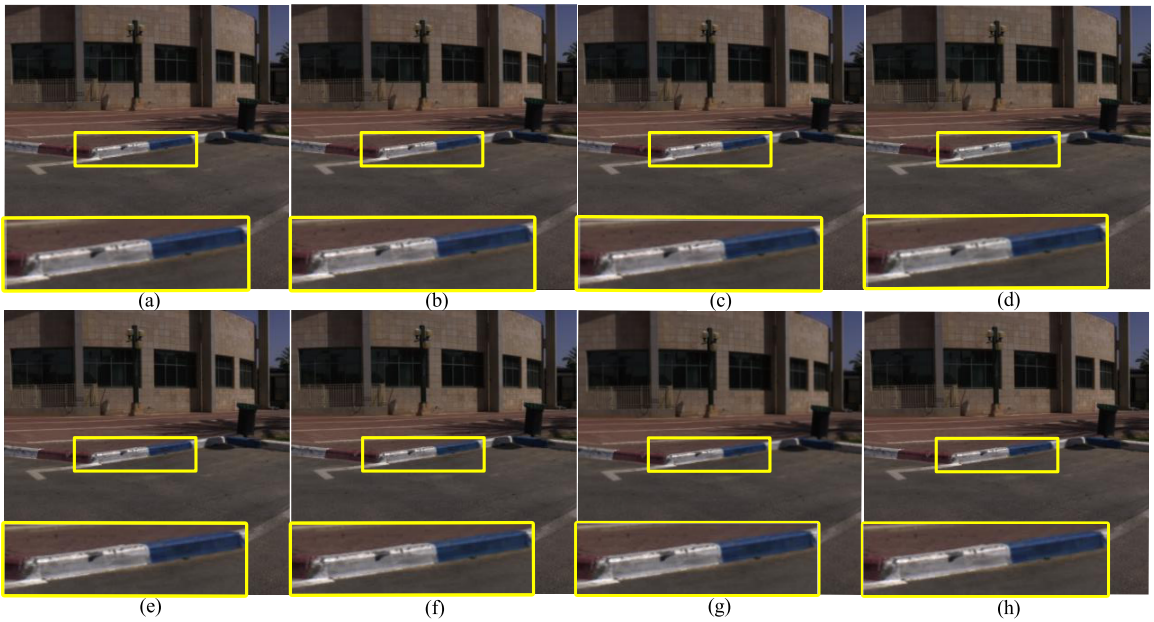


Fig. 5. False color composites (R: 600 nm, G: 560 nm, and B: 450 nm) of different fusion results. The testing image is selected from ICVL dataset with size 1024×1024, and the upscaling factor is 4. (a) Result of CNMF [16]. (b) Result of HySure [14]. (c) Result of FUSE [38]. (d) Result of Two-CNN [20]. (e) Result of DBF [24]. (f) Result of MHF-Net [25]. (g) Result of VaFuNet. (h) Ground truth.

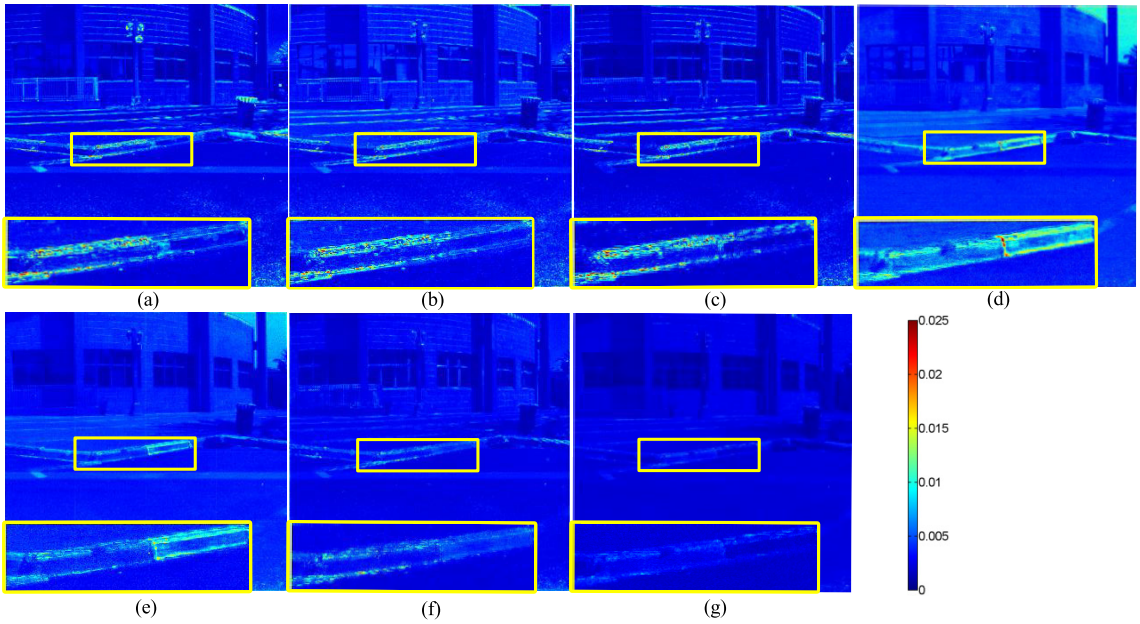


Fig. 6. Mean square error maps of different fusion results in Fig. 5. (a) Error map of CNMF [16]. (b) Error map of HySure [14]. (c) Error map of FUSE [38]. (d) Error map of Two-CNN [20]. (e) Error map of DBF [24]. (f) Error map of MHF-Net [25]. (g) Error map of VaFuNet.

factor 4, the sizes of LR HSI, HR MSI, and HR HSI samples are $10 \times 10 \times 48$, $40 \times 40 \times 4$, and $40 \times 40 \times 48$, respectively; in case of upscaling factor 8, the sizes of LR HSI, HR MSI, and HR HSI samples are $5 \times 5 \times 48$, $40 \times 40 \times 4$, and $40 \times 40 \times 48$, respectively.

In VaFuNet, the number of iterative phases K is set to 15, the number of channels per convolutional layer in the proximity operator is set to 64, and the number of pyramid levels in multiscale nonlocal attention is set to 3. The learning rate and exponential decays in the ADAM optimizer are 0.0005, 0.9, and 0.999. The batch size is set to 16. There are 15 training epochs.

B. Comparison With State-of-the-Art Methods

We compare with six state-of-the-art HSI-MSI fusion methods: CNMF method [16], hyperspectral subspace-based regularized fusion (HySure) method [14], fast fusion based on Sylvester equation (FUSE) method [38], Two-CNN method [20], multispectral–hyperspectral fusion network (MHF-Net)

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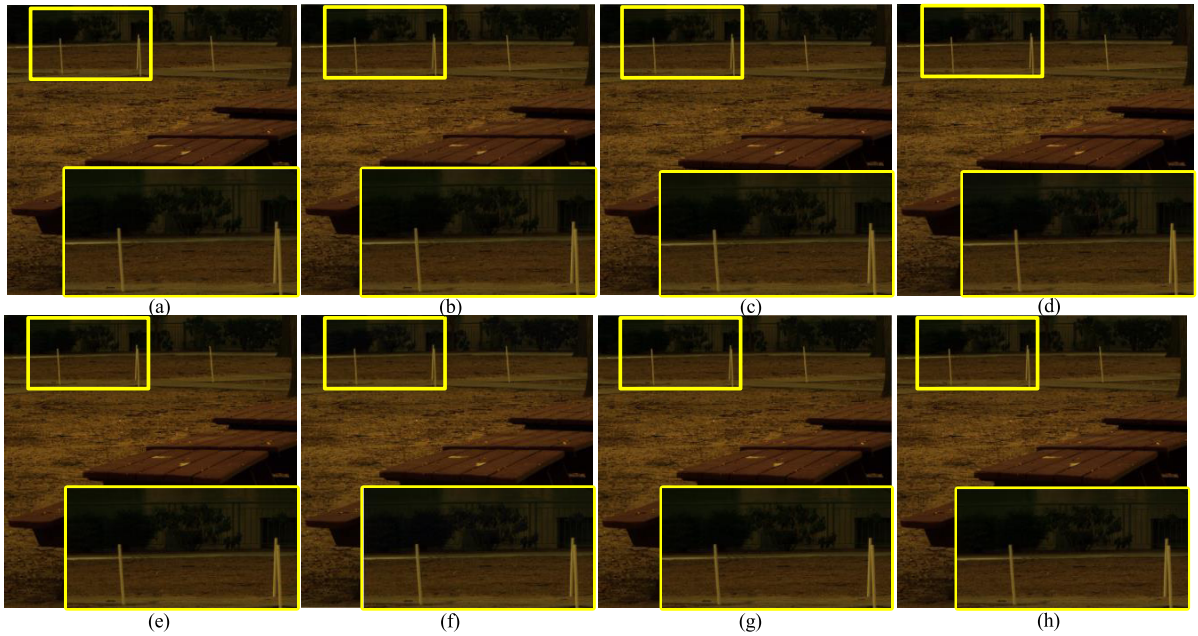


Fig. 7. False color composites (R: 660 nm, G: 540 nm, and B: 440 nm) of different fusion results. The testing image is selected from the Harvard dataset with size 1024×1024, and the upscaling factor is 8. (a) Result of CNMF [16]. (b) Result of HySure [14]. (c) Result of FUSE [38]. (d) Result of Two-CNN [20]. (e) Result of DBF [24]. (f) Result of MHF-Net [25]. (g) Result of VaFuNet. (h) Ground truth.

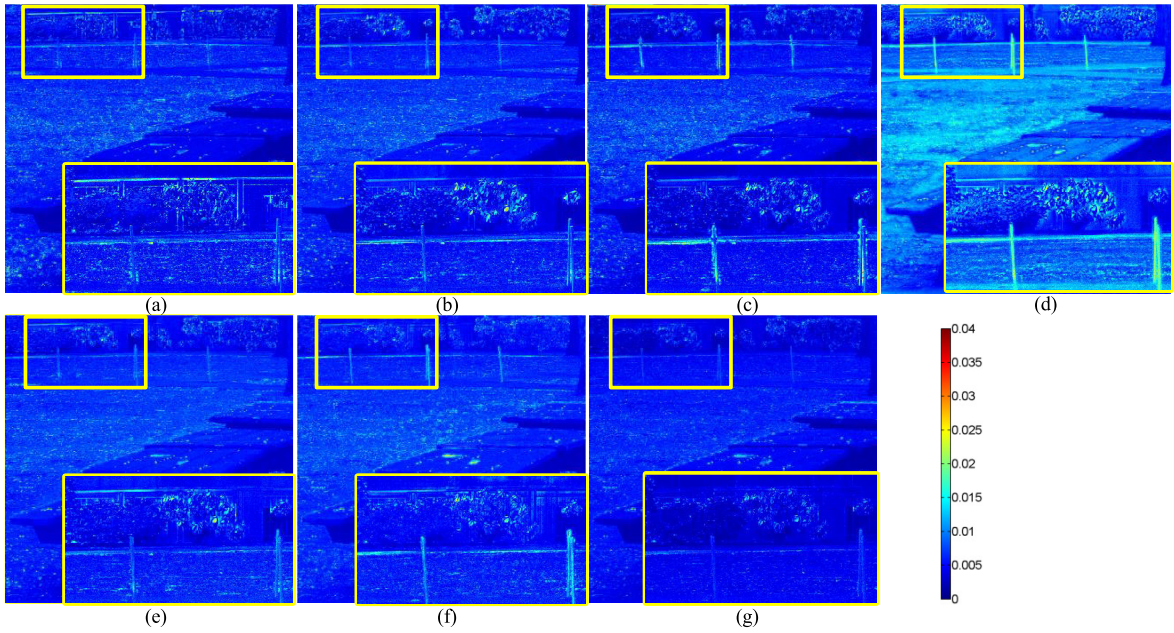


Fig. 8. Mean square error maps of different fusion results in Fig. 7. (a) Error map of CNMF [16]. (b) Error map of HySure [14]. (c) Error map of FUSE [38]. (d) Error map of Two-CNN [20]. (e) Error map of DBF [24]. (f) Error map of MHF-Net [25]. (g) Error map of VaFuNet.

method [25], and deep blind fusion (DBF) method [24]. The former three methods are model-driven and no supervised training is needed; the latter three are learning-driven and supervised learning and training samples are needed. To guarantee fairness, the number of training samples in Two-CNN, MHF-Net, and DBF is the same as VaFuNet. The reconstructed HSIs are evaluated using peak signal-noise-ratio (PSNR, dB), structural similarity index measurement (SSIM) [46], feature similarity index measurement (FSIM) [47], and mean spectral angle (MSA).

The assessment indices of different fusion methods are presented in Table I. For the ICVL and Harvard datasets, we evaluate different fusion methods on all of the 15 testing images and then compute the means of the assessment indices and report them in Table I. Both deep and multiscale nonlocal priors are utilized in our VaFuNet method. Instead of manually modeling the image prior in a handcrafted manner as in [14], VaFuNet learns the latent prior from external data and obtains a more complex prior. On the other hand, VaFuNet could jointly exploit the multiscale nonlocal prior

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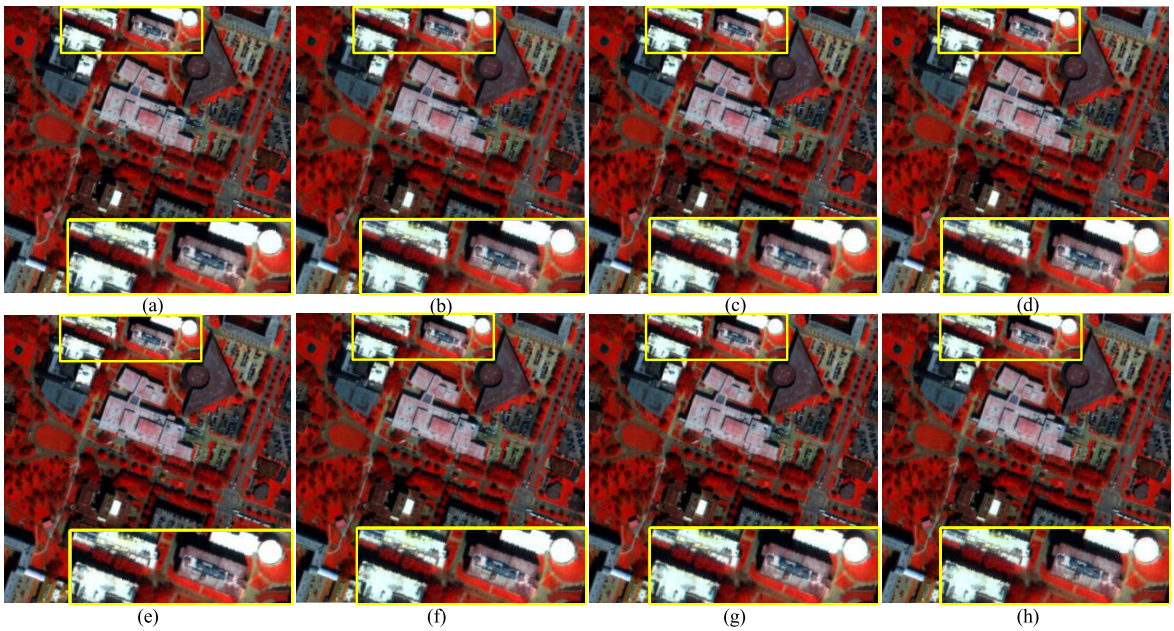


Fig. 9. False color composites (R: 1018.5 nm, G: 560.6 nm, and B: 446.1 nm) of different fusion results. The testing image is selected from *Houston University* with size 512×512, and the upscaling factor is 4. (a) Result of CNMF [16]. (b) Result of HySure [14]. (c) Result of FUSE [38]. (d) Result of Two-CNN [20]. (e) Result of DBF [24]. (f) Result of MHF-Net [25]. (g) Result of VaFuNet. (h) Ground truth.

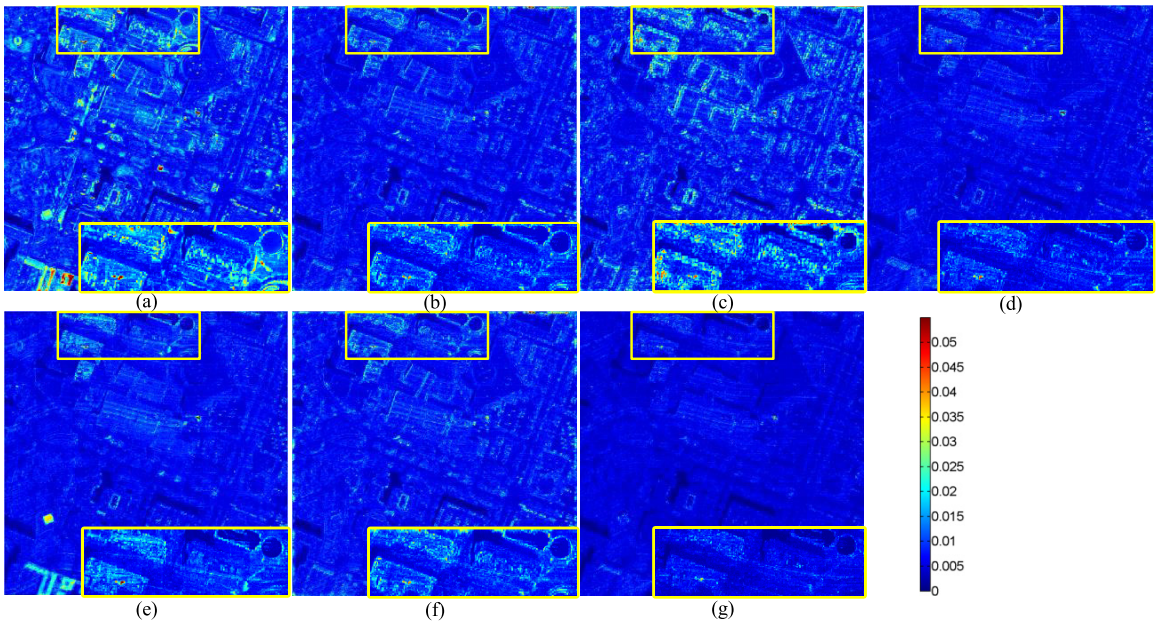


Fig. 10. Mean square error maps of different fusion results in Fig. 9. (a) Error map of CNMF [16]. (b) Error map of HySure [14]. (c) Error map of FUSE [38]. (d) Error map of Two-CNN [20]. (e) Error map of DBF [24]. (f) Error map of MHF-Net [25]. (g) Error map of VaFuNet.

and deep prior to regularize the fusion model, which is rarely considered in other DL fusion methods [19], [20]. Jointly exploiting the deep and multiscale nonlocal priors is beneficial for HSI-MSI fusion, as shown in Table I. The PSNR, SSIM, and FSIM indices of VaFuNet results are higher, and the MSA indices of VaFuNet are lower than others, which means that the proposed method can produce HR HSI with better spatial structures and less spectral distortion.

We show some of the fusion results in Figs. 5, 7, and 9. It is observed that the reconstructed HR HSIs are close to each other. To visually highlight their differences, we also give the mean square error maps over the whole bands in Figs. 6, 8, and 10, in which the reconstruction error at each pixel can be reflected. While some fusion results are close to each other, their differences in performance can be observed in the error maps. It is clear that the fusion results of VaFuNet contain lower reconstruction errors than others.

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TABLE I

ASSESSMENT INDICES OF DIFFERENT FUSION METHODS ON THE THREE HSI DATASETS

Data	r	Indices	Unsupervised (on CPU)			Supervised (on GPU)			VaFuNet
			CNMF [16]	HySure [14]	FUSE [38]	Two-CNN [20]	DBF [24]	MHF-Net [25]	
ICVL	×4	PSNR(dB)	52.3650	50.8006	51.6315	48.2279	50.7180	52.1134	**54.4341**
		SSIM	0.9945	0.9943	0.9947	0.9962	0.9965	0.9964	**0.9975**
		FSIM	0.9994	0.9991	0.9996	0.9992	0.9995	0.9998	**0.9999**
		MSA	1.6567°	1.8577°	1.4873°	2.2365°	1.5932°	1.6259°	**1.1660°**
		Time(s)	288.5	1332.1	6.9	60.5	**6.4**	25.7	7.2
	×8	PSNR(dB)	51.3802	49.6774	51.3667	45.8925	47.6818	50.0465	**53.0080**
		SSIM	0.9940	0.9933	0.9935	0.9935	0.9946	0.9952	**0.9969**
		FSIM	0.9990	0.9985	0.9989	0.9981	0.9986	0.9985	**0.9996**
		MSA	1.8655°	2.1954°	1.6985°	3.1410°	2.2887°	2.2208°	**1.3388°**
		Time(s)	241.8	1301.4	**6.3**	68.6	7.4	23.2	7.9
Harvard	×4	PSNR(dB)	41.3788	41.4017	41.6373	37.8390	39.9680	41.0318	**42.1229**
		SSIM	0.9604	0.9629	0.9627	0.9590	0.9615	0.9648	**0.9673**
		FSIM	0.9957	0.9977	0.9974	0.9939	0.9943	0.9964	**0.9994**
		MSA	4.1529°	3.8884°	3.5719°	5.3490°	4.0087°	3.6778°	**3.3293°**
		Time(s)	281.9	1232.4	7.6	62.5	**6.8**	25.8	7.1
	×8	PSNR(dB)	40.9503	40.5605	41.1442	37.2260	39.3168	39.9545	**41.3671**
		SSIM	0.9584	0.9591	0.9593	0.9572	0.9601	0.9603	**0.9628**
		FSIM	0.9933	0.9944	0.9947	0.9906	0.9922	**0.9954**	0.9951
		MSA	4.0890°	4.2741°	3.8357°	5.4554°	4.3106°	4.0629°	**3.6724°**
		Time	244	1195.0	**6.2**	64.3	6.7	22.7	7.7
Houston University	×4	PSNR(dB)	42.0859	45.4859	44.3282	47.8520	45.5055	49.3583	**50.1215**
		SSIM	0.9871	0.9910	0.9888	0.9973	0.9929	0.9982	**0.9985**
		FSIM	0.9958	0.9975	0.9956	0.9996	0.9989	0.9992	**0.9998**
		MSA	3.1944°	2.7418°	2.8926°	2.5930°	2.7227°	2.0611°	**1.7729°**
		Time(s)	54.7	205.8	**1.6**	46.7	3.4	12.1	4.8
	×8	PSNR(dB)	38.3475	43.8684	43.8414	46.8444	45.2905	48.0827	**48.5941**
		SSIM	0.9721	0.9879	0.9859	0.9964	0.9926	0.9976	**0.9980**
		FSIM	0.9913	0.9962	0.9944	0.9993	0.9987	0.9994	**0.9995**
		MSA	5.0813°	3.4096°	3.6102°	2.7081°	2.9189°	2.4388°	**2.1711°**
		Time(s)	57.9	207.6	**1.4**	47.3	3.6	11.1	4.1

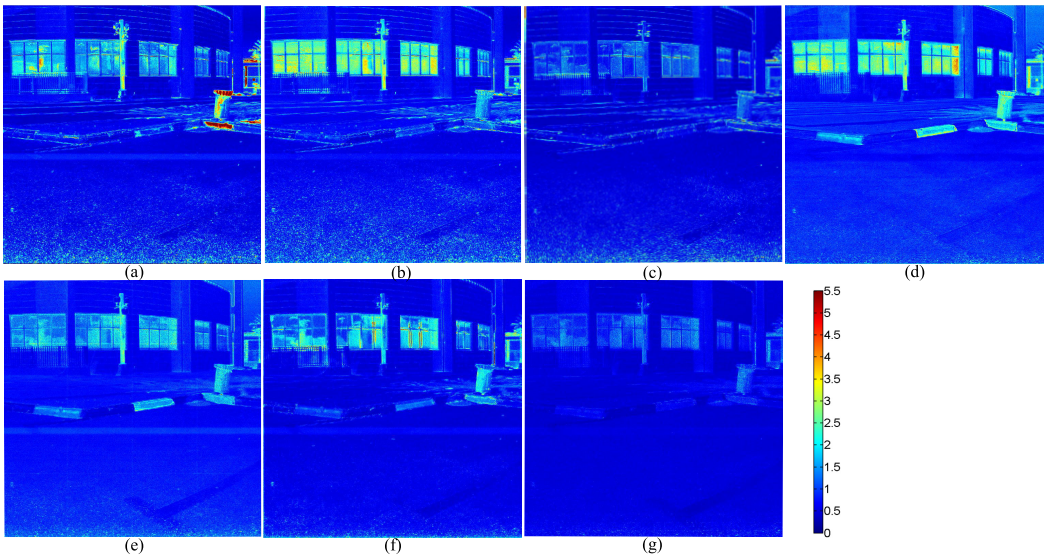


Fig. 11. Spectral angle maps of different fusion results in Fig. 5. (a) CNMF [16]. (b) HySure [14]. (c) FUSE [38]. (d) Two-CNN [20]. (e) DBF [24]. (f) MHF-Net [25]. (g) Proposed VaFuNet.

The reconstruction errors tend to occur in the regions with detailed structures, such as edges and textures, as indicated in the magnified boxes. In Figs. 6, 8, and 10, VaFuNet can reconstruct the detailed structures well with less reconstruction errors.

We also compare the running time of different fusion methods in Table I. Optimization is involved in the model-driven methods, and they generally require a longer time than learning-driven methods. Accelerated on a GPU, the learning-driven methods could reconstruct HR HSI fast. DBF is the fastest learning-driven method, and VaFuNet is slightly slower than DBF, while other assessment indices of VaFuNet are more competitive.

To further compare the spectral distortion of different fusion results, we present the spectral angle maps of the fusion results in Figs. 11–13, which reflects the spectral distortion at

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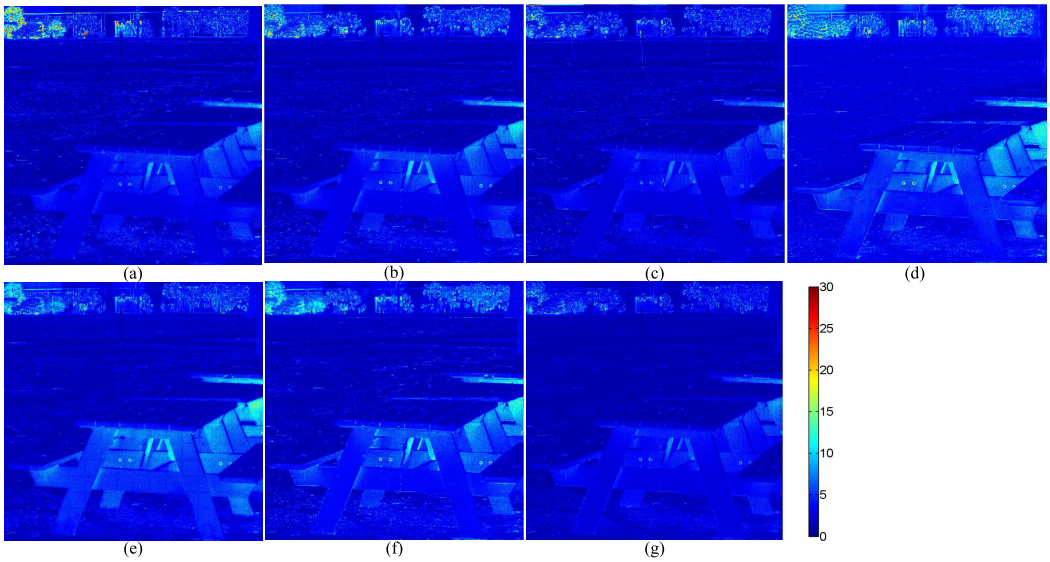


Fig. 12. Spectral angle maps of different fusion results in Fig. 7. (a) CNMF [16]. (b) HySure [14]. (c) FUSE [38]. (d) Two-CNN [20]. (e) DBF [24]. (f) MHF-Net [25]. (g) Proposed VaFuNet.

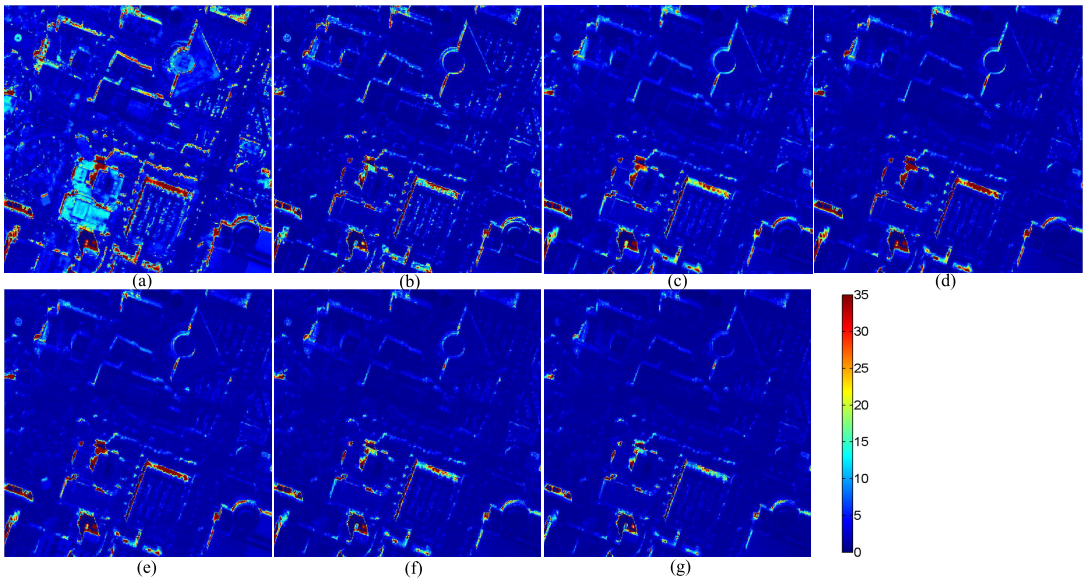


Fig. 13. Spectral angle maps of different fusion results in Fig. 9. (a) CNMF [16]. (b) HySure [14]. (c) FUSE [38]. (d) Two-CNN [20]. (e) DBF [24]. (f) MHF-Net [25]. (g) Proposed VaFuNet.

TABLE II

PSNR (dB) AND GPU RUNNING TIME (S) OF VAFUNET WITH DIFFERENT NUMBERS OF ITERATIONS PHASES AND ATTENTIVE BLOCKS PER PROXIMITY, THE UPSCALING FACTOR IS 4

(K, S)	ICVL		Harvard		*Houston University*	
	PSNR	Time	PSNR	Time	PSNR	Time
(12, 1)	53.4431	4.4	41.9566	4.1	50.0618	2.4
(15, 1)	54.4341	8.2	42.1229	7.9	50.1215	4.8
(15, 3)	54.5462	8.7	42.1619	8.2	50.3845	4.9
(1, 15)	49.3719	1.3	40.1390	1.4	49.2567	0.9

each pixel. The spectral distortion in CNMF results is higher, as shown in Table I and Figs. 11–13. It is clear that the spectral distortion of VaFuNet is generally lower than other fusion methods and closer to zero, which is beneficial for application.

C. Sensitivity Analysis of Some Network Parameters

The VaFuNet network consists of K iteration phases, each of which contains a CNN proximity composed of S multiscale nonlocal attentive CNN blocks. The setting of parameters (K, S) is important for the performance of VaFuNet. We analyze the performance of VaFuNet with different setting of (K, S), as shown in Table II. By increasing the number of iterations phases K, the network becomes deeper and the performance improves, as shown by comparing the performance between (12, 1) and (15, 1). A similar phenomenon can also be observed by increasing the attentive CNN blocks S in each phase, as shown by comparing the performance for (15, 1) and (15, 3). However, a deeper network requires more running time, and we should balance the tradeoff between the

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TABLE III

PSNR (dB) OF DIFFERENT FUSION METHODS WITH DIFFERENT SETTINGS OF GAUSSIAN DOWNSAMPLING KERNELS, THE UPSCALING FACTOR IS 4

	$\sigma = 1$			$\sigma = 3$		
	ICVL	Harvard	*Houston University*	ICVL	Harvard	*Houston University*
CNMF [16]	48.9711	42.6194	40.5915	49.4711	41.4604	39.1882
HySure [14]	48.7658	41.6203	40.4167	49.9201	41.8418	42.1156
FUSE [38]	49.1629	41.4521	39.7087	47.8825	41.1289	37.5689
Two-CNN [20]	47.3458	38.8942	46.4427	46.6084	36.7166	46.6043
DBF [24]	50.3205	40.8563	46.4964	49.7875	40.5692	46.5679
MHF-Net [25]	49.6302	41.0674	47.5685	49.0227	40.8281	46.4763
FuNet	**53.9711**	**42.6988**	**50.2034**	**52.7349**	**41.4521**	**47.8536**

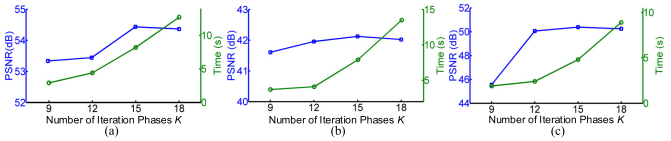


Fig. 14. Effective of number of iterations phases K on PSNR and GPU time, on (a) ICVL, (b) Harvard, and (c) *Houston University*.

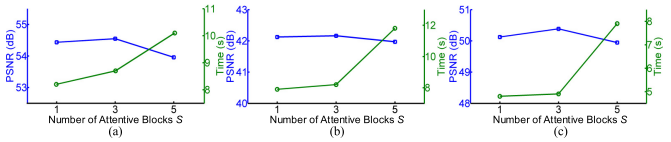


Fig. 15. Effective of number of attentive blocks S on PSNR and GPU time, on (a) ICVL, (b) Harvard, and (c) *Houston University*.

performance and the depth. It is interesting that the setting (1, 15) leads to an unexpected deficient performance even though it has the same depth as (15, 1). This can be interpreted that increasing the iteration phases of VaFuNet is more important than increasing the attentive CNN blocks per phase.

The parameters K and S are set to 15 and 1, respectively. To further investigate the effectiveness of these two parameters, we conduct more experiments and present the relationship curves between K/S and PSNR/time in Figs. 14 and 15. We fix one parameter and vary another and then observe the fusion performance. With the parameter that goes up, the inferring time increases significantly, while the PSNR would not always increase. The PSNR indices first increase and then plateau or decline slightly, and such saturation or degeneration phenomenon was also observed in other super-resolution literature works [48], which may be caused by the difficulty of training a deep network.

The Gaussian kernel is widely used in spatial downsampling because it could approximate the modulation transfer function of imaging system. Here, we investigate the robustness over the downsampling kernel by varying the standard deviation of Gaussian kernel. We take upscaling factor 4 as an example, and in this case, the default standard deviation σ of Gaussian kernel is 1.6987. When σ is set to 1 and 3, VaFuNet could still achieve competitive performance, which demonstrates the robustness of the proposed fusion method, as shown in Table III.

D. Rationality Analysis of Multiscale Nonlocal Attention

To enhance the representability and exploit the multiscale nonlocal prior of HSI, the multiscale nonlocal attention is embedded into the proximity operator of VaFuNet. In this section, we analyze the rationality behind the multiscale nonlocal attention with an ablation study.

TABLE IV

PSNR (dB) OF VAFUNET WITH DIFFERENT SETTINGS OF MULTISCALE NONLOCAL ATTENTION, THE UPSCALING FACTOR IS 4

Cases	ICVL	Harvard	*Houston University*
1	**54.4341**	**42.1229**	**50.1215**
2	53.8945	41.9913	49.8835
3	53.4372	41.7094	49.6693
4	53.2650	41.6609	49.2484
5	49.6549	38.6780	48.6886

1) *Case 1:* Channel attention and multiscale nonlocal attention are used, the number of pyramid levels is set to 3, and this is the default setting of VaFuNet.
2) *Case 2:* Channel attention and multiscale nonlocal attention are used, the number of pyramid levels is set to 1, which reduces to nonlocal attention as [49].
3) *Case 3:* Channel attention and local spatial attention are used, which reduces to the attention mechanism in [31].
4) *Case 4:* Only channel attention is used, while no spatial attention is utilized, which reduces to the channel attention in squeeze and excitation network [30].
5) *Case 5:* No attention is used, and in this case, the CNN block in each proximity reduces to a residual CNN.

In Table IV, it is clear that case 5 leads to the lowest PSNR indices, which supports the application of embedding attention mechanism in our proposed fusion model. By adopting channel attention, VaFuNet performs much better, as shown in case 4. With spatial attention introduced, VaFuNet would be further improved, as in case 3. However, the spatial attention in case 3 could only exploit the local correlation, better results can be achieved by nonlocal spatial attention exploiting the long-range interdependency of HSI, as shown in case 2. The structural recurrences exist across multiple scales, and our proposed multiscale nonlocal attention could employ the nonlocal prior from multiscale spaces, rather than single-scale space, so VaFuNet with multiscale nonlocal attention performs better than others, as shown in case 1 in Table IV.

E. Visualization of Intermediate Features

We take a testing image from the ICVL dataset as an example and visualize $\mathcal{X}^k (k = 1, 2, \ldots, K)$ generated by different iteration phases. It is noted that due to the end-to-end fashion,

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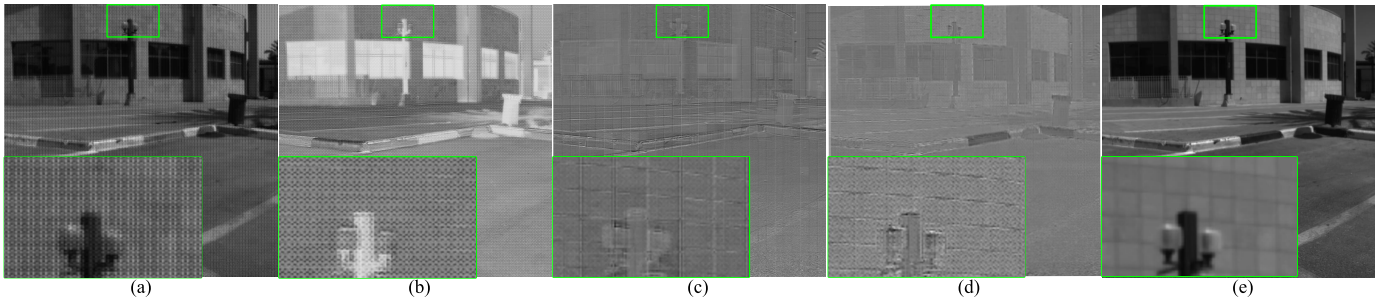


Fig. 16. Visualization of intermediate features, the 30th band of feature cubes. (a) $\mathcal{X}^1$. (b) $\mathcal{X}^5$. (c) $\mathcal{X}^{10}$. (d) $\mathcal{X}^{14}$. (e) $\mathcal{X}^{15}$.



Fig. 17. Testing image taken over Lafayette. (a) False color composite (R: 803.3 nm, G: 559.1 nm, and B: 487.8 nm) of Hyperion HSI with size 600×600. (b) False color composite (R: 835.1 nm, G: 560.0 nm, and B: 496.6 nm) of Sentinel-2 MSI with size 1800×1800.

$\mathcal{X}^K$ is the final fusion result, while $\mathcal{X}^k (k = 1, 2, \ldots, K-1)$ are actually intermediate features of the network, rather than real iterative fusion results. $\mathcal{X}^1$ is generated by bottom layers, and there are visible grid-like artifacts in $\mathcal{X}^1$, which may be caused by the transposed convolution, as shown in Fig. 16(a). With the evolution of layers, the artifacts progressively become weaker, and some detailed structures (e.g., the bricks of building) are reconstructed, as shown in Fig. 16(d). Compared with the intermediate features, the final fusion result (i.e., $\mathcal{X}^{15}$) contains less artifacts and more structural information.

F. Performance on Real-Life HSI-MSI

To investigate the applicability of the proposed fusion method, we also test different fusion methods on real spaceborne data. The HSI was collected by the Hyperion sensor carried on the Earth Observing-1 (EO-1) satellite. The MSI data were taken by the Sentinel-2 (S2) satellite. The Hyperion HSI contains 242 bands in the spectral range of 400–2500 nm with a spatial resolution of 30 m. After removing the water absorption bands and noisy bands, there are 83 bands remained. The S2 MSI has 13 bands in total, and we select four bands centered at 490, 560, 665, and 842 nm as HR MSI for fusion. The spatial resolution of these bands is 10 m.

In this real application, our goal is to fuse the 30-m Hyperion HSI and 10-m S2 MSI and produce a 10-m HR HSI. Apparently, there is no 10-m HR HSI for training. Following the strategy in [25] and [37] and Wald protocol [42], [43], we first downsample the 30-m Hyperion HSI and 10-m Sentinel MSI into 90- and 30-m resolutions by a factor of three, respectively. The choice of a factor of three is because the enhancement is from 30 m (HSI) to 10 m (fusing with MSI). To assess whether the DL fusion model is well trained, the original 30-m HSI is used as ground truth. After the model is trained, we feed the original 30-m HSI and 10-m MSI to the model and then obtain an HR HSI with 10-m resolution.

The experiment data in this real-life application were taken over Lafayette, IN, USA, and the overlapping coverage of the Hyperion and S2 images has, respectively, 2350×990 pixels and 7050×2970 pixels. First, we set aside an area with rich textures for blind test. The area represents 600×600 Hyperion pixels and 1800×1800 S2 pixels, as shown in Fig. 17. Then, image patches for training are extracted from the remainder. Again, we utilize data augmentation (rotating and flipping) and totally generate 200 000 random training samples. Each sample contains three image patches: downsampled Hyperion HSI $10 \times 10 \times 83$, S2 MSI $30 \times 30 \times 4$, and original Hyperion HSI $30 \times 30 \times 83$.

The fusion results are shown in Fig. 18. An HR HSI with 10-m resolution can be generated by fusing the 30-m Hyperion HSI and the 10-m S2 MSI. In order to highlight the details of the fusion results, we also zoom into two small areas in Fig. 18. As shown in the green boxes in Fig. 18(c) and (f), some fusion methods, such as HySure [14] and MHF-Net [25], suffer from grid-like artifacts, while such artifacts are not observed in our fusion results. In this experiment, the Hyperion HSI and the S2 MSI were not captured at the exactly same time, and the temporal difference is about one month. The change of endmembers during this month may lead to the spectral distortion. By comparing the fusion results with the original LR images, CNMF [16], Two-CNN [20], and DBF [24] tend to generate results with darker color than the original LR HSI, as shown in the yellow boxes in Fig. 18(b), (e), and (g). It means that the spectral distortion of these methods is heavier.

Because there is no HR ground-truth HSI in real-life data, in order to quantitatively evaluate the fusion results, assessment index without reference is often used [50]. We adopt the no reference quality assessment score method in [51], which measures the distortion in the spectral and spatial domains of reconstructed HSI. The lower assessment score means smaller reconstruction distortion and better fusion performance. As shown in Table V, the no-reference image

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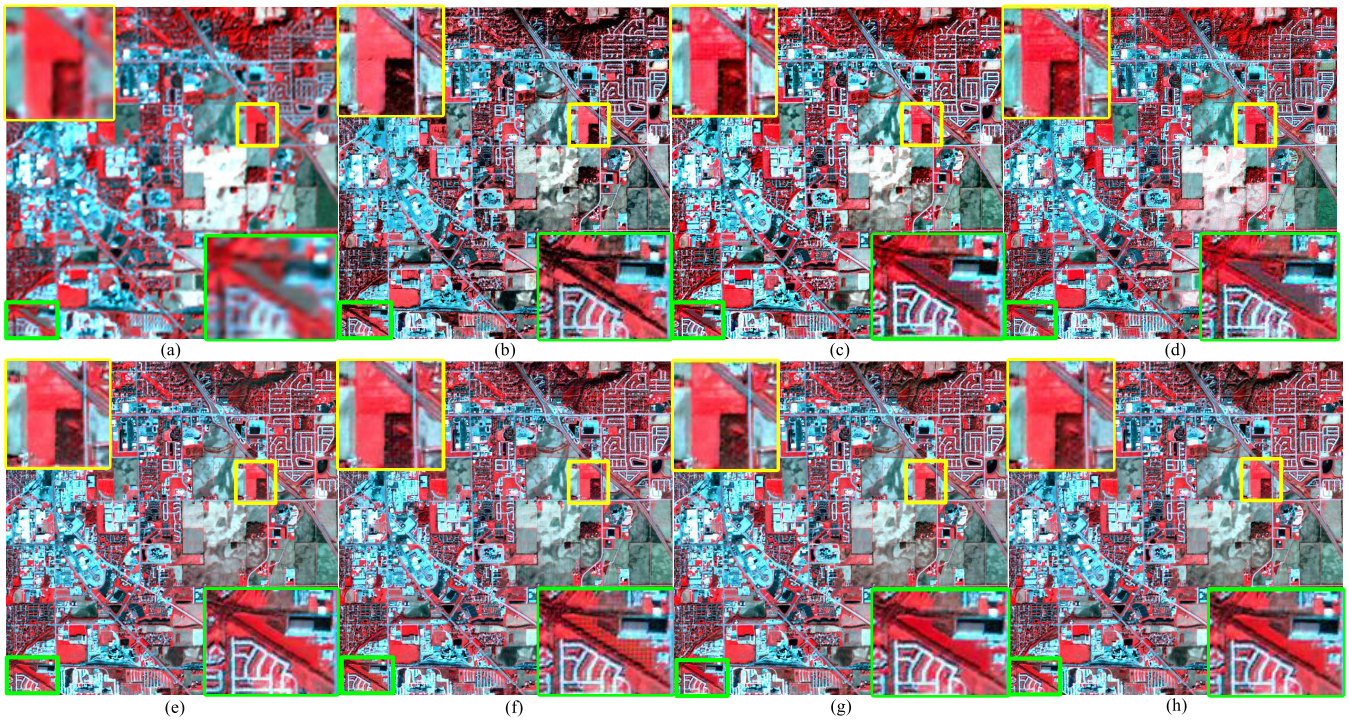


Fig. 18. False color composite (R: 803.3 nm, G: 559.1 nm, and B: 487.8 nm) of different Hyperion-Sentinel fusion results, size of the enhanced image is 1800 × 1800 with 10-m resolution. (a) Original LR HSI. (b) Result of CNMF [16]. (c) Result of HySure [14]. (d) Result of FUSE [38]. (e) Result of Two-CNN [20]. (f) Result of MHF-Net [25]. (g) Result of DBF [24]. (h) Result of our VaFuNet.

TABLE V
NO-REFERENCE IMAGE QUALITY SCORES OF DIFFERENT HYPERION-SENTINEL FUSION RESULTS

	CNMF [16]	HySure [14]	FUSE [38]	Two-CNN [20]	DBF [24]	MHF-Net [25]	VaFuNet
Scores	19.8140	18.5647	19.5758	16.6785	16.1627	15.8048	**15.7531**

quality score of VaFuNet result is competitive compared with other methods.

V. CONCLUSION

Effectively learning the degradation model and data prior is important for HSI-MSI fusion. In this study, we propose a variational deep HSI-MSI fusion model and then unfold the fusion model into a DL network, where the degradation model and data prior can be represented by the DL network and learned from the training data. To exploit the nonlocal prior of HSI, a multiscale nonlocal attentive proximity is proposed to represent the deep prior. The experiments on several HSI datasets demonstrate the effectiveness of the proposed method. It is noted that other than the nonlocal prior, some other priors of HSI, such as sparsity [52], [53] and low rankness [11], can also be integrated with deep prior, which will be investigated in the future work.

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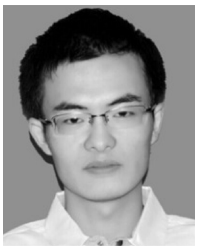
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Jingxiang Yang (Member, IEEE) received the Ph.D. degree in control theory and control engineering from Northwestern Polytechnical University, Xi'an, China, in 2019, and the Ph.D. degree in engineering science from Vrije Universiteit Brussel, Brussels, Belgium, in 2019.

He is a Lecturer with the Nanjing University of Science and Technology, Nanjing, China. His research interests include deep learning and its applications in hyperspectral image processing.

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Liang Xiao (Member, IEEE) received the B.S. degree in applied mathematics and the Ph.D. degree in computer science from the Nanjing University of Science and Technology (NJUST), Nanjing, China, in 1999 and 2004, respectively.

From 2006 to 2008, he was a Post-Doctoral Research Fellow with the Pattern Recognition Laboratory, NJUST. From 2009 to 2010, he was a Post-Doctoral Fellow with Rensselaer Polytechnic Institute, Troy, NY, USA. Since 2013, he has been the Deputy Director of the Jiangsu Key Laboratory of Spectral Imaging Intelligent Perception. Since 2014, he has been the second Director of the Key Laboratory of Intelligent Perception and Systems for High-Dimensional Information of Ministry of Education, NJUST, where he is a Professor with the School of Computer Science and Engineering. His fields of interest include remote sensing image processing, image modeling, computer vision, machine learning, and pattern recognition.



Yong-Qiang Zhao (Member, IEEE) received the B.S., M.S., and Ph.D. degrees in control science and engineering from Northwestern Polytechnical University, Xi'an, China, in 1998, 2001, and 2004, respectively.

From 2007 to 2009, he was a Postdoctoral Researcher with McMaster University, Hamilton, ON, Canada, and Temple University, Philadelphia, PA, USA. He is a Professor with Northwestern Polytechnical University. His research interests include polarization vision, hyperspectral imaging, compressive sensing, and pattern recognition.



Jonathan Cheung-Wai Chan received the Ph.D. degree from the University of Hong Kong, Hong Kong, in 1999.

From 1998 to 2001, he was with the University of Maryland, College Park, MD, USA. From 2001 to 2005, he was with the Interuniversity Micro-Electronics Centre (IMEC), Leuven, Belgium. From 2005 to 2011, he was with the Department of Geography, Vrije Universiteit Brussel (VUB), Brussels, Belgium. From 2013 to 2015, he was a Marie Curie Fellow with the Fondazione Edmund Mach, Italy. Since 2014, he has been a Guest Professor in remote sensing at the Department of Electronics and Informatics (ETRO), VUB. His research interests include machine/deep learning algorithms for classification of remotely sensed images, and spectral-spatial superresolution enhancements for spaceborne hyperspectral images.

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